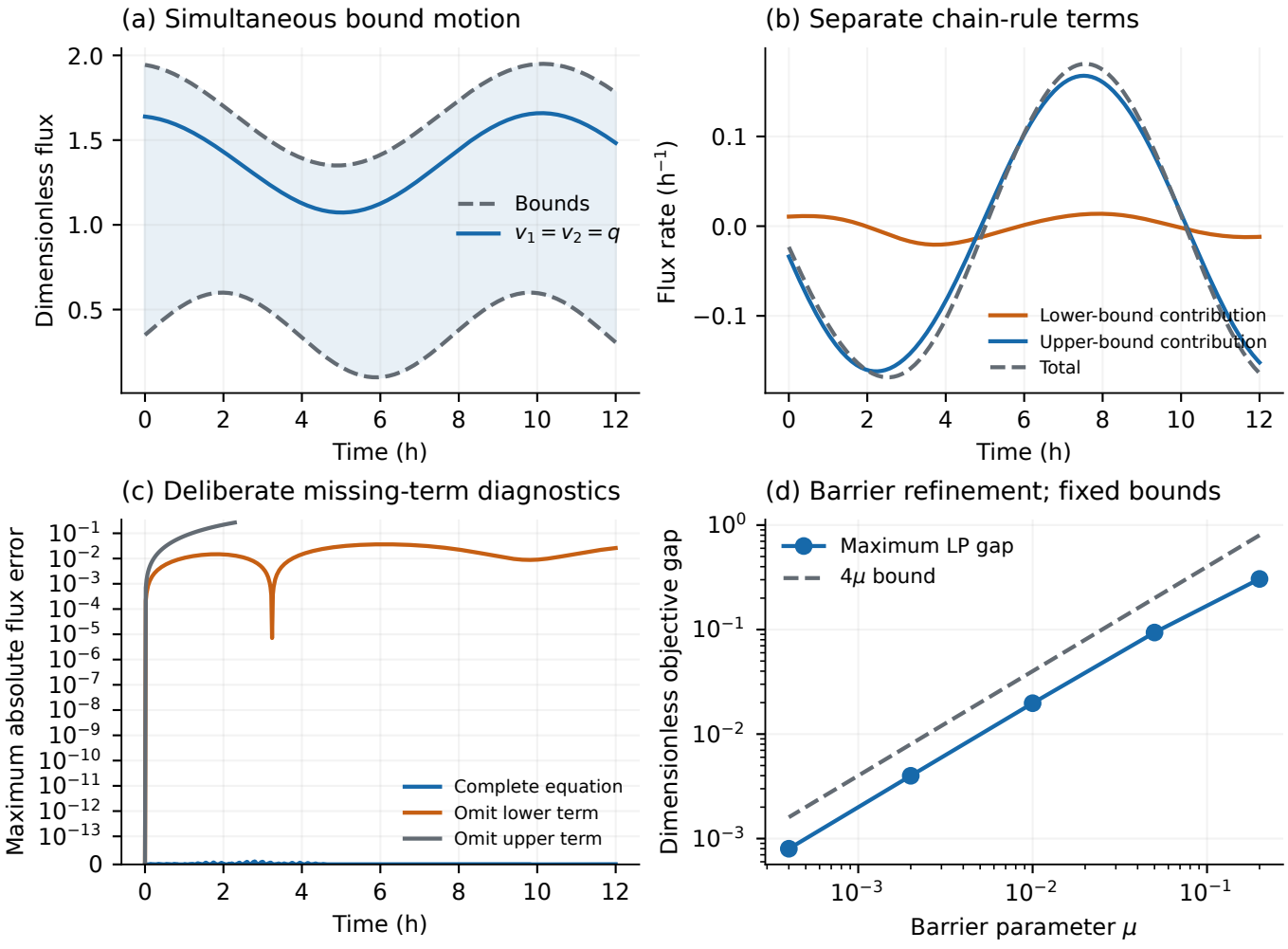
(a) Simultaneous bound motion
Dimensionless flux
Time (h)
Bounds
$v_1 = v_2 = q$
(b) Separate chain-rule terms
Flux rate (h^{-1})
Time (h)
Lower-bound contribution
Upper-bound contribution
Total
(c) Deliberate missing-term diagnostics
Maximum absolute flux error
Time (h)
Complete equation
Omit lower term
Omit upper term
(d) Barrier refinement; fixed bounds
Dimensionless objective gap
Barrier parameter μ
Maximum LP gap
4μ bound